

What I'd change

Based on driving the live system end to end with your Period 7 roster and today's published lesson (M1.T1.L1), plus an automated accessibility scan and a read of the source.

Ordered within each group by how much it costs you in a real period.

Before the critiques — the three things I'd protect. The **Screen Studio's** four-surface authoring is the best-designed thing here. The **Growth screen pre-loading planned misconceptions** with a ready teacher move, before any student data exists, is something I have not seen in a shipping product. And the **Concrete → Representational → Abstract fade actually fades** — state 8 says *"use the structure you learned from the area model, but do not reopen the grid."* Most curricula claim CRA and never remove the scaffold. Don't let any refactor eat these.

STUDENT SIDE

What an 11-year-old actually experiences on the Chromebook.

COSTLY An unverified student is invisible and their screen never moves

I joined with the code, sat on the warm-up card, and advanced the room through seven states. The student screen never changed once. Session page read **Joined: 0 of 11** the entire time, and the control panel top bar showed no waiting pill — because the pending indicator only fires when the student taps "Warm-up not connecting? Ask for help."

Your 1:1 Chromebook setup makes the common cause of this rare, which downgrades it. But the failure is silent, and silent failures are the expensive kind at 35 desks.

Fix: have code entry itself write a pending row, so the top bar reads something like `9 verified · 3 started · 23 not in`. Same UI you already built, different trigger.

COSTLY Students can't see where they are in the lesson

Your control panel knows it's Step 7 of 15 with 5:00 left and "Equation Only — Independent" up next. The student's screen knows none of that. They see the current instruction and nothing else — no position, no time remaining, no what's-coming.

For a 6th grader this is the difference between "we're in the third of four things and then I get to work with paper" and "an adult is making the screen change." Position is regulation. It's also the cheapest anxiety reduction available to you, and it's already in your data model.

Fix: a thin progress strip on the student surface — step position, the timer they can already hear, and the name of what's next.

COSTLY There is no way for a student to say "I'm confused"

This is the one that struck me hardest, because your entire stated culture is *confusion is step one, that's how you know you're engaged*. A student's only signalling affordance in the whole system is a link labelled **"Warm-up not connecting? Ask for help"** — a technical support request.

You've built a classroom philosophy around naming confusion out loud, and then built software where the only thing a kid can report is that the internet is broken.

Fix: a persistent, low-stakes tap on every student screen — "I'm stuck" / "say that again" / "I've got this." Feed it straight to the teacher surface as a live count and a name list. That single control does more for your stated pedagogy than any manipulative you could add, and it turns confusion from a thing kids hide into a thing they report.

WORTH DOING Dead air with no floor

The holding screen says "Keep this page open" with no time estimate and nothing to do. Every second of unassigned screen time on a 6th-grade Chromebook is a classroom management event, not a UX event. You own a whole practice-games library at `/practice` that never appears when a student is waiting.

Fix: after ~20 seconds of waiting, drop in a 60-second fluency drill or an Abbie card. Interrupt it the instant the teacher's state changes.

WORTH DOING Students never see their own success criteria

The lesson has a real success criterion — *"I can use an area model to write and explain two equivalent expressions for the same area."* It appears on the projector at state 3 for two minutes and then vanishes for the rest of the period. The student surface never carries it.

Self-assessment against a visible criterion is the whole point of writing one. At state 9 you ask them to rate their confidence against a criterion they last saw twenty minutes and six states ago.

WORTH DOING The absent-student path is buried

"**ABSENT STUDENT START HERE**" on the lesson page is a genuinely good idea — most products have no answer for the kid who missed Tuesday — and it's currently formatted as a small label mid-page. It should be a card, and it should be reachable from the home screen without a code, which is what `/explore` already does but doesn't advertise.

SMALLER No student-visible record of their own year

You collect warm-up scores, checkpoint results, tool evidence and mastery by domain — and a student can see none of it. Growth is visible to you and invisible to the person doing the growing. A simple "here's what you've gotten better at" view would carry your culture further than most things on this list.

TEACHER SIDE

What it's like to run the period from behind the laptop.

COSTLY Three navigation systems and duplicated controls with different names

Still the biggest structural problem. `/teacher` is a card launcher with a search bar. `/session` and every `/teacher/*` page share a 14-item cream nav. `/control` has a different 14-item dark nav. The same actions carry different labels depending on where you're standing, and — I hit this during the run — the *same* action carries different labels depending on lesson state: the advance control is **Next state** on the session page, **Advance** on the control panel, and silently becomes **Show results** whenever a response window is open.

That last one is good design in isolation. Across three surfaces it means there is no single thing you can teach your hands to do.

Fix: one IA, three modes — Prep / Run / Review. One name per action, everywhere.

COSTLY No read on whether you're ahead or behind

Fifteen states, 55 minutes, and the panel tells you the current timer and what's next — but never the thing you actually need at minute 30, which is *am I going to make it*. There's no elapsed-vs-planned, no projected end time, no marker for which states are compressible.

The states sum to roughly the full period with no slack, and 15 transitions at even 20 seconds each is five unbudgeted minutes. On a normal day with one interruption you will be improvising which state to cut, with no information about which cut costs least.

Fix: a projected-finish readout, and let each state carry a "protected / compressible / skippable" flag set in the Studio. When you're 6 minutes down, the panel tells you what to drop.

COSTLY The 55-state bank is on screen while class is running

Mid-lesson you need maybe four controls. The Run surface currently shows a 14-item nav, the state, the timer, eight transport buttons, the full lineup with per-item reorder and delete, and then the entire state bank across eight labelled categories, expanded. The highest-cognitive-load screen in the system is the one you look at with 35 kids waiting.

WORTH DOING No live view of who is on what

During Individual Work Stations — the longest block in the lesson at 14 minutes — you have no screen showing which student is at which station, who's opened the Area Tool, who hasn't touched anything. Day Review tells you afterwards. The value is during.

WORTH DOING End Session arms and silently disarms

It's a two-tap confirm, which is right. But the armed state expires with no visual countdown, so a hesitation reads as a dead button. I misdiagnosed it as broken on first pass, and I was reading the network traffic — a teacher mid-period won't be.

WORTH DOING Roster hygiene and stale displays

- **6 students are sitting in "Unassigned"** from the Notion sync with no prompt to place them. They belong to nobody and appear in no period's data.

- `/weekly-display` shows **Friday July 24** while the active lesson is dated the 25th. It's a projector screen, so it's wrong in front of the room.

SMALLER Audio lives on one laptop

Music-per-state is one of your best features and it's stored in one browser's IndexedDB, labelled "stored on this classroom laptop." It doesn't follow you to another room, doesn't survive a profile reset, and can't be shared. Cloud-store it.

ACCESSIBILITY

Automated WCAG 2.1 A/AA scan on eight student-facing pages, plus design-level issues the scanner can't catch.

COSTLY Timed auto-advance with no student-side control

This is the accessibility issue specific to your design, and no scanner will flag it. Your readiness checks run **57 seconds** and auto-submit. The teacher has pause, ± 1 minute and +30s. **The student has nothing** — no pause, no extension, no warning that submission is imminent.

For a student with a processing-speed accommodation, an IEP extended-time provision, or an English learner reading a word problem, a hard 57-second window that closes automatically is a barrier that produces bad placement data — and this is the exact item that routes them into a 14-minute differentiated station. The kid who needs the most time gets misclassified by the timer.

Fix: a per-student time multiplier on the roster ($1.5\times$ / $2\times$) that extends response windows for flagged students without changing the room's clock, plus a visible "10 seconds left" warning on the student surface. This is also just good instruction, not only compliance.

COSTLY Contrast failures on every page scanned

PAGE	SERIOUS CONTRAST VIOLATIONS	TAP TARGETS < 32PX
Home	1	1
Explore	4	1

Lesson	4	0
Practice	4	1
GEMS	6	4
Fraction Bars	2	9
Number Line	1	0
Area Model	5	4

The cream palette is an asset — it's warmer and far easier on a projector than the blue-white everyone else uses. The failure is the muted grey body text on cream, which is a one- or two-token change, not a redesign.

WORTH DOING Tap targets in the tools kids actually use

Nine sub-32px controls in Fraction Bars, four each in GEMS and Area Model. These are the manipulatives you route students into during work stations, pressed by 11-year-olds on Chromebook trackpads. 44px is the target; this is a usability fix that happens to also be a compliance fix.

WORTH DOING Audio cues have no visual equivalent for some students

Your 30-second warning, countdown tick and end tone carry real instructional information. A deaf or hard-of-hearing student gets none of it. The on-screen timer partly covers this on surfaces that show one — but the student surface during work states often doesn't, and a student wearing headphones for a different accommodation is in the same position.

Fix: mirror every sound cue as a visual state change on the student screen — a border pulse at 30s, a visible countdown from 10.

UNVERIFIED Keyboard and screen-reader operability of the manipulatives

I did not test this and I'm not going to assert it. But drag-and-tap tools like Algebra Tiles, Fraction Bars and the Area Model are the classic place where keyboard operability is absent, and the Area Model's grid inputs need programmatic labels for a screen reader to be usable.

Worth an actual pass with a keyboard only and with VoiceOver before you'd ever claim conformance.

SMALLER

Minor markup issues

- One image missing alt text on the lesson page, one on the area model page.
- The GEMS step badges overflow their circles at phone width ("MULTIPLY / DIVIDE" spills outside the ring).
- Home page scrolls 48px horizontally at 390px.

PEDAGOGY

Critique of the instructional design of M1.T1.L1 as it's encoded in the 15 states — not of your teaching, which I haven't seen. Read this as a colleague arguing about a lesson plan.

THE BIG ONE

Your stated philosophy isn't in the lesson

"Being confused is step one — that's how you know you're engaged. Step two is what do you know. Step three is try something."

Now the actual sequence: **Hook (3 min) → Learning Intention and Success Criterion (2 min) → Area Tool I Do (4 min) → We Do (4 min) → Guided Try (4 min) → Independent.**

The target is revealed at minute five, and the teacher demonstrates at minute eight. There is no window in which a student is productively stuck before being shown. That is a clean, competent gradual-release lesson — I Do / We Do / You Do — and gradual release is close to the *opposite* of confusion-first. It front-loads clarity precisely so the student never has to sit in not-knowing.

Both are defensible. But you can't run a culture built on "confusion is step one" inside a lesson structure designed to prevent confusion, and expect students to believe the poster.

What I'd try: make state 2 a genuine struggle task — give them 5×27 with grid paper and no method, three minutes, "try something" — and move the Learning Intention to *after* it, where it names what they just bumped into. Same total time. The I Do then answers a question they've already asked instead of pre-empting it.

THE BIG ONE The weakest signal is collected first, and the strongest evidence isn't used at all

State 9 is Fist-to-Five — *self-reported confidence* — and it runs **before** the two readiness checks that actually measure anything. Self-assessment is the least reliable data you will collect all period, and it is least reliable exactly for the students the routing is meant to catch: the ones who don't know what they don't know rate themselves high, and the anxious capable ones rate themselves low.

Meanwhile at state 7 every student completes an area model *and* a full equivalent-expression chain in the Area Tool — which logs their work — and none of that appears to feed the placement decision. You are routing on two multiple-choice items while sitting on a constructed response.

What I'd try: move Fist-to-Five after the checks and reframe it as reflection rather than data. Then let the state-7 tool evidence be the third input to routing. Two binary MC items give you four buckets with real classification error; adding one constructed response changes that materially at zero time cost.

WORTH ARGUING ABOUT Fourteen minutes of differentiated work on a two-question placement

Check 1 is recognition (*which equation matches the model*). Check 2 is error repair (*which revision correctly repairs $4(5 + 10) = 20 + 10$*) — and that one is genuinely well-built, because the distractor encodes exactly the **distributes-to-one-addend** misconception you pre-registered in Growth. Assessment aligned to a named misconception is the strongest design move in this lesson.

But both are multiple choice, both run 57 seconds, and together they place a student into a 14-minute station. A guess lands a struggling kid in an advanced route; one misread lands a fluent kid in remediation. Neither error is visible to you until the block is over.

What I'd try: keep both items, add a mid-block pulse. At the 7-minute mark of state 13, a 30-second "show me where you are" that lets you re-route anyone misplaced. Right now the placement is a one-way door.

WORTH ARGUING ABOUT The lesson reveals answers and skips the misconception

Both readiness checks end by displaying the correct choice — $5(20 + 7) = 5 \times 20 + 5 \times 7$ and $4(5 + 10) = 20 + 40$. That's it. The wrong answer in check 2 is *the* classic distributive error and you designed it deliberately, and then the moment passes without anyone asking why it's attractive.

This is the single clearest gap between the lesson and the philosophy. "How to think, not what to think" lives precisely in the thirty seconds after a wrong answer appears.

What I'd try: a reveal that shows the correct answer *and* asks one question — "why would someone write $20 + 10$?" Ten seconds of turn-and-talk. You already own a DiscussionProtocol component that's unused in this lesson.

WORTH ARGUING ABOUT **Fifty-five minutes with zero student-to-student talk**

Your state bank contains Discussion, Gallery Walk, Small Group Rotations, Partner / Group Work, and Error-Analysis Discussion. Today's lesson uses none of them. Every one of the 15 states is either teacher-to-class or student-alone.

For a lesson about *explaining* equivalence — the success criterion literally says "write **and explain**" — that's a mismatch. Explanation is a social act; you're assessing it in silence.

WORTH ARGUING ABOUT **No formative check between We Do and Independent**

States 5 (We Do, 4 min) → 6 (Guided Try, 4 min) → 7 (Independent, 5 min) → 8 (Equation Only, 4 min). A student who lost the thread at We Do does **nine minutes** of independent practice building the wrong procedure before any check fires at state 10. Practising an error to fluency is worse than not practising.

What I'd try: one 30-second check between 6 and 7. You have the response infrastructure already — it's the same mechanism as the readiness checks, just moved earlier.

WORTH ARGUING ABOUT **The lesson never closes its own loop**

The Learning Intention appears at state 3 and never returns. State 15 is Cleanup — logistics, not closure. There's no moment where the class revisits the target and names what changed.

Notably, your **L2 sequence does this right**: state 11 is "Learning Intention, Success Criterion, and Fist-to-Five" as a deliberate revisit. Today's lesson dropped it. Whatever made you include it in L2 is the right instinct.

CHECK THIS Warm-up may not match your own rule

Your stated warm-up format is **2 review + 3 current**, with the misconception on the third current question being the point. The student-facing instruction says "Complete all five Google Form questions" — five is right, but I couldn't see inside the form to confirm the 2+3 split or that the misconception item is doing its job. Worth checking that your own format survived contact with the form builder.

Also: the warm-up is collected and, in today's sequence, never discussed with students. L2 has a "Warm-Up Review" state at position 2. L1 doesn't. Collecting misconception data and never showing students the misconception is a missed loop.

STRUCTURAL Timing has no slack and no priorities

The 15 states sum to essentially the whole period. Fifteen transitions at 15–30 seconds each is four to seven minutes that aren't in the plan. Add one announcement, one late arrival, one tech hiccup and you're cutting states live with no guidance about which ones are load-bearing.

The fix is the same one from the teacher section: mark each state protected / compressible / skippable at authoring time, so the decision is made when you're calm rather than at minute 38.

Evidence: full drive-through of the live deployment on Period 7 (session **D0G6SD** , lesson M1.T1.L1, all 15 states), student join flow instrumented at the network level, automated WCAG 2.1 A/AA scan of 8 student-facing pages, source review at HEAD, and Notion lesson search. Pedagogy critique is of the lesson as encoded in its state definitions and screen text — I have not observed the lesson being taught, and the difference between those two things is usually most of the lesson.